

Math100C - WK9

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1.

Determine the Galois group of the splitting field of the minimal polynomials of the following quantities:

$$\sqrt{5 + \sqrt{13}}, \quad \sqrt{5 + \sqrt{21}}, \quad \sqrt[3]{2 + \sqrt{2}}, \quad \sqrt[3]{2 + \sqrt[3]{2}}$$

.....
I need so much help!

2.**(a)**

Show that for any prime p ,

$$\sum_{j=0}^{p-1} e^{2\pi i j^2 / p}$$

is a square root of either p or $-p$. Then deduce that every quadratic extension of $\mathbb{Q}$ is contained in $\mathbb{Q}(\zeta_n)$ for some positive integer n .

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Let

$$\chi(k) = \begin{cases} 1 & \text{if } k \text{ is a square mod } p \\ -1 & \text{else} \end{cases}$$

be the Legendre symbol (recall that the Legendre symbol is a homomorphism). For brevity, let ζ be the primitive p th root of unity.

$$\begin{aligned} \sum_{j=0}^{p-1} e^{2\pi i j^2 / p} &= 1 + \sum_{j \in \mathbb{F}_p^\times} \zeta^{j^2 \pmod p} \\ &= 1 + \sum_{k \in \mathbb{F}_p^\times} (1 + \chi(k)) \zeta^k \end{aligned} \tag{1}$$

$$\begin{aligned} &= 1 + \sum_{k \in \mathbb{F}_p^\times} \zeta^k + \sum_{k \in \mathbb{F}_p^\times} \chi(k) \zeta^k \\ &= \sum_{k \in \mathbb{F}_p^\times} \chi(k) \zeta^k \end{aligned} \tag{2}$$

where (1) follows because if $k \in \mathbb{F}_p$ is a square mod p , i.e., $x^2 - k$ has at least one solution in $\mathbb{F}_p$, then in fact $x^2 - k$ has exactly two solutions. Following on from (2),

$$\begin{aligned} \left(\sum_{j=0}^{p-1} e^{2\pi i j^2/p}\right)^2 &= \sum_{a \in \mathbb{F}_p^\times} \sum_{b \in \mathbb{F}_p^\times} \chi(a)\chi(b)\zeta^{a+b} = \sum_{a \in \mathbb{F}_p^\times} \sum_{b \in \mathbb{F}_p^\times} \chi(ab)\zeta^{a+b} \\ &= \sum_{a \in \mathbb{F}_p^\times} \sum_{b \in \mathbb{F}_p^\times} \chi(aab)\zeta^{a+ab}, \end{aligned} \quad (3)$$

$$\begin{aligned} &= \sum_{a \in \mathbb{F}_p^\times} \sum_{b \in \mathbb{F}_p^\times} \chi(b)\zeta^{a(1+b)} = \sum_{b \in \mathbb{F}_p^\times} \chi(b) \sum_{a \in \mathbb{F}_p^\times} \zeta^{a(1+b)} \\ &= \sum_{\substack{b \in \mathbb{F}_p^\times \\ b \neq -1}} \chi(-1) \sum_{a \in \mathbb{F}_p^\times} \zeta^0 + \sum_{\substack{b \in \mathbb{F}_p^\times \\ b \neq -1}} \chi(b) \sum_{a \in \mathbb{F}_p^\times} \zeta^{a(1+b)} \\ &= (p-1)\chi(-1) + \sum_{\substack{b \in \mathbb{F}_p^\times \\ b \neq -1}} \chi(b) \sum_{a \in \mathbb{F}_p^\times} \zeta^a \\ &= (p-1)\chi(-1) - \sum_{\substack{b \in \mathbb{F}_p^\times \\ b \neq -1}} \chi(b) \\ &= (p-1)\chi(-1) + \chi(-1) = p\chi(-1) \end{aligned} \quad (4)$$

where (3) follows from the substitution $b \mapsto ab$ and the fact that ab and b permute over the same elements in the group $\mathbb{F}_p^\times$; similarly, (4) follows from the substitution $a \mapsto a(1+b)^{-1}$ in $\mathbb{F}_p^\times$, where $(1+b)^{-1}$ is guaranteed to exist when summing over $b \in \mathbb{F}_p^\times \setminus \{-1\}$. The equality

$$\left(\sum_{j=0}^{p-1} e^{2\pi i j^2/p}\right)^2 = p\chi(-1)$$

shows that $(\sum_{j=0}^{p-1} e^{2\pi i j^2/p})^2$ is indeed a square root of p or $-p$.

(b)

Show every quadratic extension of $\mathbb{Q}$ is contained in some $\mathbb{Q}(\zeta_n)$.

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All quadratic extensions of $\mathbb{Q}$ are of the form $\mathbb{Q}(\sqrt{d})$ for a square-free integer d . Write $d = p_1 \cdots p_k$ with each prime factor distinct. By subsection (a),

$$\sqrt{d} \in \mathbb{Q}(\zeta_{p_1}, \dots, \zeta_{p_k}, i) = \mathbb{Q}(\zeta_n)$$

where $i = \sqrt{-1}$ is thrown in since, a priori, it's unclear if which of $\sqrt{+d}$ or $\sqrt{-d}$ is inside $\mathbb{Q}(\zeta_{p_1}, \dots, \zeta_{p_k})$. This shows $\mathbb{Q}(\sqrt{d}) \subset \mathbb{Q}(\zeta_n)$.

3.

Let $f(x) = x^5 - 3x + 1$.

(a)

Show f is irreducible over $\mathbb{Q}$.

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 By the rational root theorem or whatever, only possible roots of f are ± 1 ; clearly neither are roots, so f has no linear factor. Suppose $f(x) = (x^2 + ax \pm 1)(x^3 + bx^2 + cx \pm 1)$. In the $+1$ case, one can solve the linear system to get

$$c - (3 + c)^2 + 1 = 0,$$

which has no solutions over $\mathbb{R}$ (and thereby $\mathbb{Q}$). Similarly, in the -1 case, one obtains

$$3 - a - a^2 = 0$$

but a cannot be rational since $3 - x - x^2$ is irreducible over $\mathbb{Q}$ by the rational root test. Hence f cannot have a quadratic root either; f is irreducible over $\mathbb{Q}$.

(b)

Verify using some computer system to show $f(x)$ has exactly three roots.

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 Graph the function on desmos! The three real roots of f are around -1.4 , 0.3 , and 1.2 .

(c)

Deduce that the Galois group of f is $\mathfrak{S}_5$.

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 Since f has complex roots (which must be conjugates), the Galois group of f contains a transposition; namely, the automorphism that swaps the complex roots and is identity on all other elements. Further, since f is irreducible, the Galois group contains a 5-cycle. Hence the Galois group of f is $\mathfrak{S}_5$.

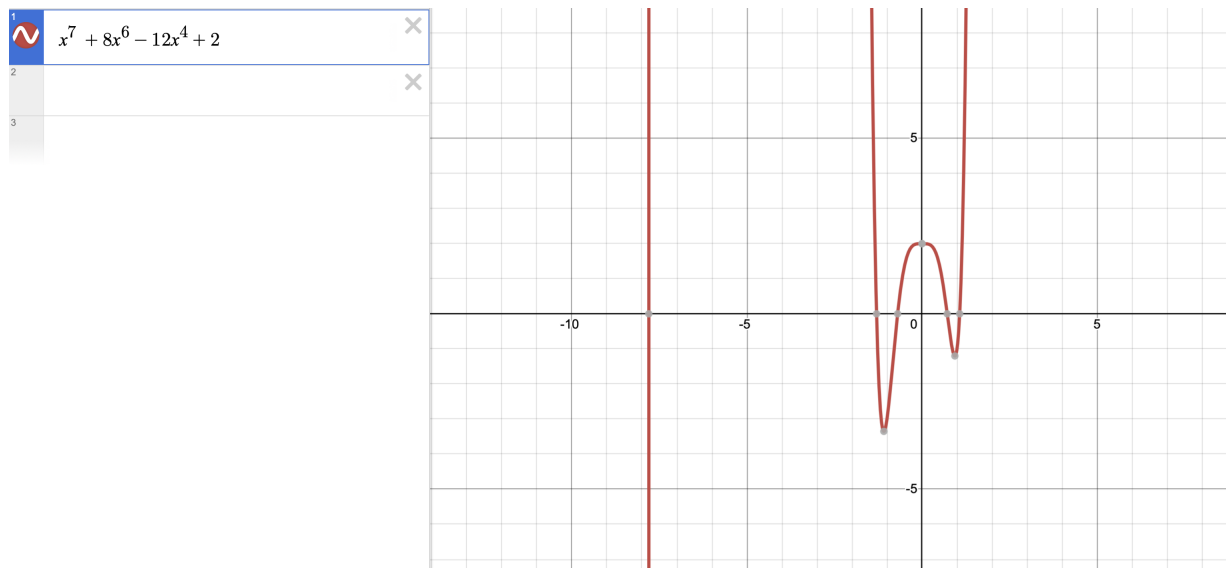
4.

Find a degree 7 polynomial whose Galois group is $\mathfrak{S}_7$.

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$$f(x) = x^7 = 8x^6 - 12x^4 + 2$$

satisfies the desired property. It is irreducible by Eisenstein for $p = 2$, and it has five real roots:



By the same process as Q3, the Galois group of f is $\mathfrak{S}_7$.

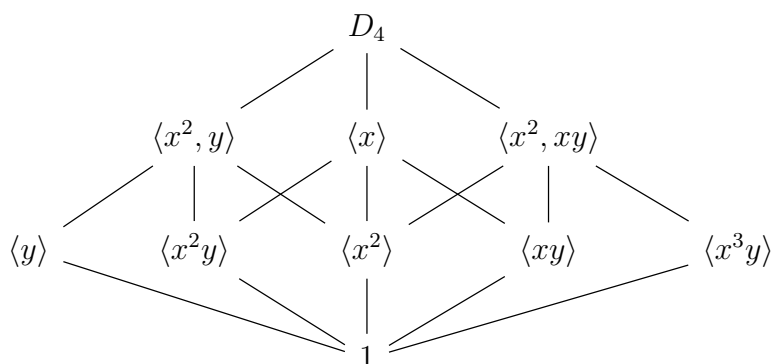
5.

Let K/F be a Galois extension. Determine the number of subfields of K of each possible degree over F in each of the following cases:

$$\text{Gal}(K/F) \cong D_4, \quad \text{Gal}(K/F) \cong A_4$$

(a)

Let $D_4 = \langle x, y \mid x^4 = y^2 = 1; yx = x^3y \rangle$. The diagram below is the lattice for D_4 : it contains two copies of the Klein Four group, one 4-cycle, and five 2-cycles.

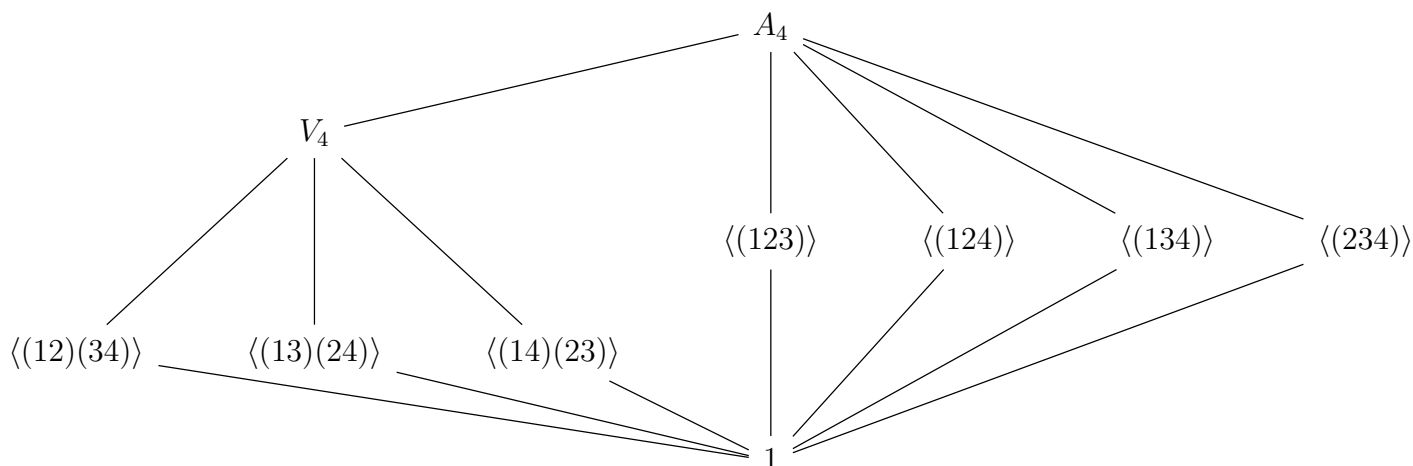


Accordingly, by the main theorem of Galois theory, the subfields of K over F are:

- One degree 8 extension, namely K itself
- Five degree 4 extensions corresponding to the 2-cycles

- Three degree 2 extensions corresponding to the Klein Four groups
- One degree 1 extension, F itself

(b)



The above is the lattice for A_4 : clearly its order 2 and order 3 groups are generated by the double transpositions and three-cycles. Since there are only three nonidentity elements of A_4 (double transpositions) with order dividing 4, the only possible order four subgroup would be the set containing the identity and all the double transpositions, and it turns out this is the Klein Four group. The only other possible subgroup of A_4 is a subgroup of order 6, but not such subgroup exists: any subset of A_4 with six elements has at least one three-cycle and a double transposition, but then a subgroup generated by a three-cycle and a double-transposition is the entire A_4 .

Accordingly, by the main theorem of Galois theory, the subfields of K over F are:

- One degree 12 extension, K itself
- Three degree 6 extension, corresponding to the two-cycles
- Four degree 4 extensions, corresponding to the three-cycles
- One degree 3 extension, corresponding to V_4
- One degree 1 extension, F itself

6.

Show that A_6 is simple.

I saw this proof in Rotman's *An Introduction to the Theory of Groups*. Take a nontrivial normal subgroup $H \triangleleft A_6$ and a nontrivial element $h \in H$. Consider the case that some index k is fixed by h . Letting

$$K = \{a \in A_6 \mid a(i) = i\},$$

K can be identified as A_5 since it consists of even permutations of $6 - 1 = 5$ indices. $h \in H \cap K$, so namely $H \cap K$ is nontrivial. But clearly $H \cap K \triangleleft K$; A_5 is simple, so one concludes $H \cap K = K$, i.e., $K \subset H$. K contains a three-cycle of A_5 , so H contains a three-cycle, say (123) . But all three-cycles are conjugate in A_6 : take another three-cycle (xyz) . A permutation $\sigma \in \mathfrak{S}_6$ sending

$$1 \mapsto x, \quad 2 \mapsto y, \quad 3 \mapsto z$$

satisfies $\sigma(123)\sigma^{-1} = (abc)$. Namely, σ can be taken to be even: if some σ mapping as above is odd, then multiply σ by (45) . Then conjugating (123) by $\sigma(45)$, an even permutation, still yields (xyz) . Hence H contains all the 3-cycles, and the 3-cycles generate A_6 . Hence $H = A_6$.

In other words, if $H \triangleleft A_6$ contains any element that fixes at least one index, then $H = A_6$. Next, consider the case that no element of H fixes any index. After relabeling, the elements of $\mathfrak{S}_6$ that achieve this are of the form

$$(12)(34)(56), \quad (12)(3456), \quad (123)(456)$$

but only the latter two are even permutations. So all nonidentity elements of H are of those two cycle types. But the second power of $(12)(3456)$ is a four cycle that fixes 1 and 2. Meanwhile, since H is normal in A_5 , if H contains $(123)(456)$, then it contains

$$(234)(123)(456)(243) = (14)(265)$$

which is an odd permutation, a contradiction. This shows that the only normal, nontrivial subgroup of A_6 is A_6 itself.

7.

Show that every Galois extension of degree 10 is solvable.

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Let G be the Galois group of a degree 10 Galois extension. Notice $10 = 2 \cdot 5$ is a group of order pq with $q \mid (p - 1)$. By the Sylow theorems, there are two isomorphism classes for G : $G \cong Z_2 \times Z_5 \cong Z_{10}$, the cyclic group of order 10, or $G \cong Z_2 \rtimes Z_5 \cong D_5$, the dihedral group of a pentagon. Any abelian group is solvable, and the chain of subgroups of $D_5 = \langle x, y \mid x^5 = y^2 = 1; yx = x^4y \rangle$

$$1 < \langle x \rangle < D_5,$$

with the identification $D_5/\langle x \rangle \cong \langle y \rangle \cong Z_2$, shows D_5 is a solvable group. Hence any Galois extension of degree 10 is solvable.

8.

(a)

I follow the derivation given by Kiran. Let $f = x^p - a$ and ζ be the primitive p th root of unity. Let K be the splitting field for f over F . The quotient of the roots of f is a p th root of unity, so $F(\zeta) \subset K$.

First, assume this inclusion is proper. By Kummer's theorem, the intermediate extension $K/F(\zeta)$ is cyclic of degree p . But then the Galois group of this intermediate extension is a p -cycle on the roots of f , i.e., it is transitive on the roots of f . This shows f is irreducible.

It follows that if f factors, then $F(\zeta) = K$. The Galois group of a Galois cyclotomic extension is a subgroup of $(\mathbb{Z}/p\mathbb{Z})^\times$. Call the Galois group $G = \langle c \rangle \subset (\mathbb{Z}/p\mathbb{Z})^\times$; it has order d dividing $p - 1$. Viewing c as both the number generating $(\mathbb{Z}/p\mathbb{Z})^\times$ and the automorphism generating the Galois group,

$$c : \zeta \mapsto \zeta^c$$

If $c = 1$, then the Galois group is trivial; $K = F$ and f splits in F , and there is nothing more to show. Suppose $c \neq 1$. Take β to be an arbitrary root of f .

$$c : \beta \mapsto \zeta^k \beta,$$

for some $k \in \mathbb{Z}/p\mathbb{Z}$, so the orbit of β under the action of $G = \langle c \rangle$ is

$$\beta, \zeta^k \beta, \zeta^{ck+k} \beta, \zeta^{c^2k+ck+k} \beta, \dots, \quad (5)$$

where the elements eventually cycle. If $k = 0$, then the orbit is just β and $\beta \in F$ since β is fixed by every element of G . Suppose $k \neq 0$. Disregarding the factor k , notice that exponents of ζ are

$$0, 1, c + 1, (c^2 + c + 1), \dots$$

i.e., of the form $(c^t - 1)/(c - 1)$. By definition of $G = \langle c \rangle$, c has order d in $(\mathbb{Z}/p\mathbb{Z})^\times$, so that general term has form $(c^t - 1)/(c - 1)$ shows the cycle above has period d , i.e., (5) has d distinct elements.

The paragraph above shows that in the case $K = F(\zeta)$ with $[K : F] = d$, the orbits of the roots under the action of the Galois group has sizes 1 or d . Suppose $d > 1$ and none of the orbits of the roots has size 1. Then $p = d \cdot (\text{number of orbits})$, i.e., $d \mid p$. But by definition of d , $d \mid p - 1$. A number greater than 1 cannot simultaneously divide p and $p - 1$, hence a contradiction. Consequently, at least one root of f is fixed by the entire Galois group, i.e., F contains at least one root of f , as was to be shown.

(c)

As Kiran suggested, consider $p = 2$ and $a = -4$. $x^2 + 4$ is clearly irreducible over $\mathbb{Q}$ since it has no rational (or even real) roots. However,

$$x^4 + 4 = (x^2 - 2x + 2)(x^2 + 2x + 2),$$

shows $x^4 + 4$ is not irreducible. This gives a counterexample to the proposition for $p = 2$.

Acknowledgements

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